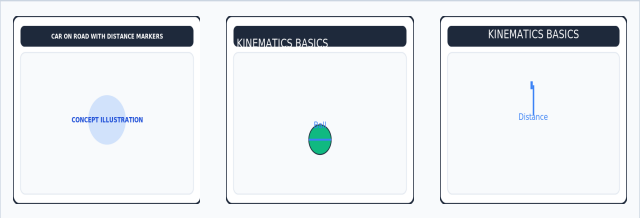
DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework



Kinematics Basics

Subject: Physics | **Grade:** 11

Introduction to Kinematics

Main Concept: Study of motion without considering forces

Motion is movement of an object from one place to another. Position tells where object is. Time tells when it happens.

Key Points:

• Define motion

• Describe position

• Explain time

Real-World Example: Car driving from Pokhara to Kathmandu

Key Vocabulary:

• **motion:** change in position over time

Types of Motion

Main Concept: Different types of motion exist

Objects can move in a straight line, circle, or curve.

Key Points:

• Straight line

• Circular motion

• Curved path

Real-World Example: Ball rolling on flat surface

Key Vocabulary:

• **straight line:** linear motion

Measuring Distance

Main Concept: Distance is measured

Distance is how far an object travels. Use tools like rulers or count steps.

Key Points:

• Use ruler

• Measure path

• Count steps

Real-World Example: Measuring distance between two trees

Key Vocabulary:

• **distance:** length of path traveled

Calculating Speed

Main Concept: Speed is calculated

Speed is distance divided by time. Use formula to find speed.

Key Points:

• Use formula

• Divide distance

• Time taken

Real-World Example: Calculating speed of a car

Key Vocabulary:

• **speed:** rate of motion

Real-life Examples

Main Concept: See motion in everyday life

We see motion in walking, running, and driving.

Key Points:

• Walking

• Running

• Driving

Real-World Example: People walking in Tundikhel

Key Vocabulary:

• **real-life:** everyday examples

Review Kinematics

Main Concept: Review key concepts

Review key concepts of motion, types, distance, and speed.

Key Points:

• Define motion

• Types of motion

• Measuring distance

• Calculating speed

Real-World Example: Recap with questions

Key Vocabulary:

• **motion:** change in position over time

• **distance:** length of path traveled

• **speed:** rate of motion
